

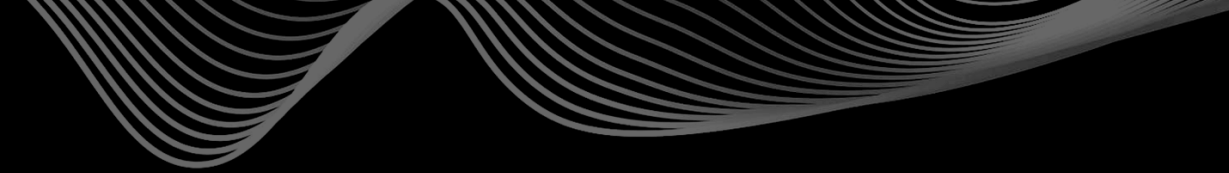
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GLOBAL FASHION RETAIL ANALYTICS

**END-TO-END DATA ANALYTICS & MACHINE LEARNING
PROJECT**

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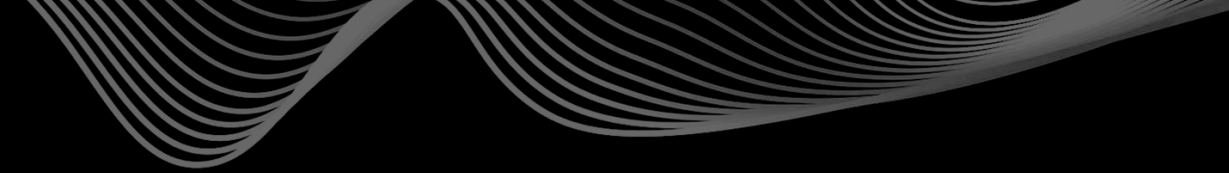
PROJECT OVERVIEW

PURPOSE OF THE PROJECT

- TRANSFORM RAW RETAIL DATA INTO ACTIONABLE INSIGHTS
- SUPPORT BUSINESS STRATEGY WITH ANALYTICS
- BUILD SQL → POWER BI → MACHINE LEARNING PIPELINE

WHAT WE DELIVER

- ADVANCED SQL INSIGHTS
- INTERACTIVE DASHBOARDS
- PREDICTIVE ML MODELS



PROBLEM STATEMENT

BUSINESS CHALLENGES

- OPTIMIZING PRODUCT ASSORTMENT & PRICING
- IDENTIFYING HIGH-VALUE CUSTOMERS
- IMPROVING PROMOTION EFFICIENCY
- REDUCING RETURNS
- IMPROVING OPERATIONAL EFFICIENCY

GOAL:

CONVERT DATA INTO INSIGHTS THAT SUPPORT STRATEGIC
DECISION-MAKING.

PART 1

SQL ANALYSIS OVERVIEW

SQL INSIGHTS

- CUSTOMER ANALYTICS
- SALES VS RETURNS
- CATEGORY PERFORMANCE
- PROMOTIONS
- STORE COMPARISON
- REVENUE & GROWTH TRENDS

FOR SQL QUERIES PLEASE REFER BELOW LINK:

[Global Fashion Retail SQL Analysis.sql](#)

1. Retrieve the number of unique customers by country and city along with their average purchase value.

Explanation:

- **Count(Distinct c.Customer_ID)**
Counts unique customers per country + city.
- **Avg(t.Line_Total)**
Returns the average amount spent per line item for customers in that location.
- **Left Join**
Ensures cities with customers but no transactions still appear.

###Retrieve the number of unique customers by country and city along with their average purchase value.

```
Select
    c.Country,
    c.City,
    count(distinct c.Customer_ID) as unique_customers,
    Avg(t.Line_Total) as avg_purchase_value
From customers c
Left Join transactions t
    On c.Customer_ID = t.Customer_ID
Group By
    c.Country,
    c.City
Order by
    c.Country,
    c.City;
```

Country	City	unique_customers	avg_purchase_value
Deutschland	Ahrensburg	1423	39.256469133483385
Deutschland	Altona	1550	40.34485034486405
Deutschland	Bad Homburg	964	40.82372536433963
Deutschland	Barsinghausen	94	46.940434663192086
Deutschland	Bergedorf	1502	40.85137784407836
Deutschland	Bergisch Gladbach	1186	40.91353073441956
Deutschland	Berlin	29000	40.01923217452609
Deutschland	Böblingen	822	44.359511647555394
Deutschland	Bochum	506	40.80315320556228
Deutschland	Böhlen	268	46.312598495032844
Deutschland	Bornheim	1091	39.87103550206424
Deutschland	Brandenburg an ...	1581	40.20944987062659
Deutschland	Brühl	1170	39.3887327064253
Deutschland	Castrop-Rauxel	427	39.4457576491616
Deutschland	Charlottenburg	1692	40.136220179163104
Deutschland	Darmstadt	980	40.622288525011726
Deutschland	Delitzsch	205	49.56628205226018
Deutschland	Dortmund	522	41.070810829196965
Deutschland	Düsseldorf	696	51.17543659210205
Deutschland	Eimsbüttel	1518	41.91885523564824
Deutschland	Erfstadt	1059	40.15504890937211
Deutschland	Erkrath	632	42.47177542810855
Deutschland	Eschborn	901	37.54234232666256
Deutschland	Esslingen	854	42.21841967787989
Deutschland	Fellbach	806	43.56486803485501

2. Calculate the percentage of returns compared to total transactions and track how it changes monthly.

Explanation:

- `Date_format(Date, '%Y-%m')` → groups transactions monthly (YYYY-MM)
- `count(*)` → total number of transactions (sales + returns)
- `sum(case when Transaction_Type = 'return' then 1 end)` → number of returns
- `return_percentage` → $(\text{returns} \div \text{total transactions}) \times 100$

```
#Calculate the percentage of returns compared to total transactions and track how it changes monthly.
Select
    date_format(t.Date, '%Y-%m') as yearmonth,
    count(*) as total_transactions,
    sum(case when t.Transaction_Type = 'Return' then 1 else 0 end) as return_transactions,
    round(
        (sum(case when t.Transaction_Type = 'Return' then 1 else 0 end)
         / count(*)) * 100,
        2
    ) as return_percentage
from transactions t
group by DATE_FORMAT(t.Date, '%Y-%m')
order by yearmonth;
```

yearmonth	total_transactions	return_transactions	return_percentage
2023-01	171759	9256	5.39
2023-02	119380	6501	5.45
2023-03	264005	14322	5.42
2023-04	187136	9986	5.34
2023-05	213135	11215	5.26
2023-06	132241	6893	5.21
2023-07	147591	7725	5.23
2023-08	128408	6705	5.22
2023-09	309401	16253	5.25
2023-10	273977	14098	5.15
2023-11	196629	10358	5.27
2023-12	665851	35450	5.32
2024-01	178489	9453	5.30
2024-02	144528	7652	5.29
2024-03	316217	16421	5.19
2024-04	203406	10679	5.25
2024-05	226064	11850	5.24
2024-06	156722	8416	5.37
2024-07	152775	8132	5.32
2024-08	158834	8396	5.29
2024-09	358683	19318	5.39
2024-10	301067	15758	5.23
2024-11	223236	12018	5.38
2024-12	730940	38592	5.28
2025-01	176889	9515	5.38

3. Identify the top categories by total revenue and their share in overall sales.

Explanation:

- Join products on Product_ID: Correct relationship to link each transaction to its product category.
- Sum(Line_Total): Computes revenue per category.
- Cross join total: Provides the overall sales total for percentage calculation.
- $(\text{category revenue} / \text{total revenue}) \times 100$: Gives each category's contribution to total sales.
- Results sorted by highest revenue category first.

```
# Identify the top categories by total revenue and their share in overall sales.
Select
    p.Category,
    sum(t.Line_Total) as category_revenue,
    round(
        (sum(t.Line_Total) / total.total_revenue) * 100,
        2
    ) as revenue_share_percentage
from transactions t
join products p
    on t.Product_ID = p.Product_ID
cross join (
    select SUM(Line_Total) as total_revenue
    from transactions
) as total
group by p.Category, total.total_revenue
order by category_revenue desc;
```

	Category	category_revenue	revenue_share_percentage
►	Feminine	340685345.8434937	46.49
	Masculine	326979394.612859	44.62
	Children	65080134.56652975	8.88

4. Compare total sales volume and average basket value across payment methods.

Explanation:

- Total sales volume = SUM(Line_Total), total revenue generated by that payment method
- Average basket value = AVG(Invoice_Total), average value of a complete invoice
- Grouped by payment method = so we can compare card vs cash vs others

```
##Compare total sales volume and average basket value across payment methods.  
Select  
    t.Payment_Method,  
    sum(t.Line_Total) as total_sales_volume,  
    avg(t.Invoice_Total) as avg_basket_value  
from transactions t  
Where t.Transaction_Type = 'sale'  
Group by t.Payment_Method  
Order by total_sales_volume desc;
```

Payment_Method	total_sales_volume	avg_basket_value
Credit Card	621514010.4102228	274.6653405880162
Cash	154546512.24460554	273.11924796137475

5. Compare sales volume and revenue during discount periods versus non-discount periods.

Explanation:

- Case When t.Discount > 0 then 'Discount Period' else 'Non-Discount Period' end as period_type
- Categorizes each sale as either a discount or non-discount sale.
- count(*) as total_transactions: Counts total number of transactions in each period type.
- sum(t.Line_Total) as total_revenue: Calculates total revenue for each period type.
- where t.Transaction_Type = 'sale': Excludes returns from the calculation.
- Group by period_type: Groups results into discount vs non-discount periods.

```
##Compare sales volume and revenue during discount periods versus non-discount periods.  
Select  
    case  
        when t.Discount > 0 then 'Discount Period'  
        else 'Non-Discount Period'  
    end as period_type,  
    count(*) as total_transactions,  
    sum(t.Line_Total) as total_revenue  
from transactions t  
where t.Transaction_Type = 'sale'  
group by period_type;
```

period_type	total_transactions	total_revenue
Non-Discount Period	4288334	631904692.1977148
Discount Period	1788866	144155830.4571135

6. Measure the proportion of new versus repeat customers each month along with their average spend.

Explanation:

CTE 1

- Identifies the earliest purchase month for each customer.
- Converts all dates into the first day of their month (e.g., “2024-02-15” to “2024-02-01”), ensuring month-level comparison. It establishes each customer’s month - the month they first became a customer.

CTE 2

- Aggregates purchasing behavior at the customer × month level.
- For each customer and transaction month, it computes the total spend for that month.
- This forms the basis for determining whether a customer is new or repeat each month.

Final calculations:

- Counts customers whose current month is their first purchase month.

Continue on next page

```
with first_purchase as (  
    select  
        Customer_ID,  
        min(date_format(date, '%Y-%m-01')) as First_Purchase_Month  
    from transactions  
    group by Customer_ID  
)  
customer_monthly as (  
    select  
        Customer_ID,  
        date_format(date, '%Y-%m-01') as Txn_Month,  
        sum(Invoice_Total) as Monthly_Spend  
    from transactions  
    group by Customer_ID, date_format(date, '%Y-%m-01')  
)  
select  
    cm.Txn_Month as month,  
    sum(case when cm.Txn_Month = fp.First_Purchase_Month then 1 else 0 end) as New_Customers,  
    sum(case when cm.Txn_Month > fp.First_Purchase_Month then 1 else 0 end) as Repeat_Customers,  
    sum(case when cm.Txn_Month = fp.First_Purchase_Month then 1 else 0 end)  
        / count(distinct cm.Customer_ID) as New_Proportion,  
    sum(case when cm.Txn_Month > fp.First_Purchase_Month then 1 else 0 end)  
        / count(distinct cm.Customer_ID) as Repeat_Proportion,  
    avg(case when cm.Txn_Month = fp.First_Purchase_Month then cm.Monthly_Spend end) as Avg_Spend_New,  
    avg(case when cm.Txn_Month > fp.First_Purchase_Month then cm.Monthly_Spend end) as Avg_Spend_Repeat  
from customer_monthly cm  
join first_purchase fp using (Customer_ID)  
group by cm.Txn_Month  
order by cm.Txn_Month;
```

6. Measure the proportion of new versus repeat customers each month along with their average spend.

Explanation:

Continue from previous page:

- Counts customers buying after their first purchase month.
- calculates proportions by dividing counts by the total number of unique customers active that month:
- The average monthly spend of new customers and average monthly spend of repeat customers.
- Finally group by txn_months and orderby the same.

month	New_Customers	Repeat_Customers	New_Proportion	Repeat_Proportion	Avg_Spend_New	Avg_Spend_Repeat
2023-01-01	110456	0	1.0000	0.0000	266.10993880635283	NULL
2023-02-01	68575	9485	0.8785	0.1215	319.573878545556	264.8394527775147
2023-03-01	134769	29144	0.8222	0.1778	392.8159699469723	407.4353028879596
2023-04-01	83311	36317	0.6964	0.3036	419.9270504495205	423.7763716541474
2023-05-01	85055	50180	0.6289	0.3711	378.15934749807565	392.64171897235167
2023-06-01	47897	38465	0.5546	0.4454	448.4304236173456	457.5581450664035
2023-07-01	49742	46258	0.5181	0.4819	375.6397169386251	382.5291744054299
2023-08-01	39920	44196	0.4746	0.5254	409.9182715127965	413.2264967808001
2023-09-01	86904	102634	0.4585	0.5415	369.27176349191916	393.01164536289645
2023-10-01	66982	102509	0.3952	0.6048	442.23042903981985	462.28030566115405
2023-11-01	43030	81997	0.3442	0.6558	389.9165418948846	408.1771089669729
2023-12-01	122957	242163	0.3368	0.6632	300.6543954390559	345.1680411541446
2024-01-01	29640	85347	0.2578	0.7422	316.4257700022898	313.56605734590426
2024-02-01	21323	72645	0.2269	0.7731	359.2530483578068	370.2881950550787
2024-03-01	44338	148833	0.2295	0.7705	388.0667823978483	411.61821192451913
2024-04-01	25991	104235	0.1996	0.8004	481.4719325920511	513.3027208686441
2024-05-01	27195	115715	0.1903	0.8097	373.02077022032785	413.94239984225607
2024-06-01	17113	84125	0.1690	0.8310	471.8743353006486	510.1129594065537
2024-07-01	15917	83220	0.1606	0.8394	422.6976251485038	453.44588440981255
2024-08-01	15629	87279	0.1519	0.8481	384.4994400661031	442.34482235969097
2024-09-01	32050	183481	0.1487	0.8513	383.9830714510235	450.6329698694409
2024-10-01	24293	160462	0.1315	0.8685	415.46291087216764	504.79389980051474
2024-11-01	16745	124452	0.1186	0.8814	406.9275246218519	478.7779336456413
2024-12-01	48418	345531	0.1229	0.8771	296.65088396877195	393.9358001165519

7. Identify the top and bottom stores by revenue and return rates.

Explanation:

CTE 1: store_stats

- Joins each store with its transactions
- Calculates total revenue (sum of sales)
- Counts how many transactions were returns
- Counts all transactions
- Computes the return rate as: return transactions/ total transactions
- Produces one row per store with all relevant KPIs
- This forms the statistical base for ranking stores.

CTE 2: Rank Stores by Performance

- row_number() assigns 1 to the best/worst store for each metric.Example:

Top_Revenue_Rank = 1 i.e. Store with the highest revenue

Bottom_Revenue_Rank = 1 i.e. Store with the lowest revenue

Top_Return_Rank = 1 i.e. Store with the highest return rate

Continue on next page

```
with store_stats as (  
    select  
        s.Store_ID,  
        s.Store_Name,  
        s.City,  
        s.Country,  
        sum(case when t.Transaction_Type = 'Sale' then t.Invoice_Total else 0 end) as Revenue,  
        sum(case when t.Transaction_Type = 'Return' then 1 else 0 end) as Return_Count,  
        count(*) as Total_Transactions,  
        sum(case when t.Transaction_Type = 'Return' then 1 else 0 end) * 1.0 / count(*) as Return_Rate  
    from stores s  
    left join transactions t on s.Store_ID = t.Store_ID  
    group by s.Store_ID, s.Store_Name, s.City, s.Country  
)  
ranked as (  
    select  
        *,  
        row_number() over (order by Revenue desc) as Top_Revenue_Rank,  
        row_number() over (order by Revenue asc) as Bottom_Revenue_Rank,  
        row_number() over (order by Return_Rate desc) as Top_Return_Rank,  
        row_number() over (order by Return_Rate asc) as Bottom_Return_Rank  
    from store_stats  
)
```

7. Identify the top and bottom stores by revenue and return rates.

Explanation:

Continue from previous page:

Bottom_Return_Rank = 1 i.e. Store with the lowest return rate

This allows selecting top/bottom performers without multiple queries or UNIONS.

- Selects only the stores ranked #1 in each category
- Labels each store:
 - Top Revenue
 - Bottom Revenue
 - Top Return Rate
 - Bottom Return Rate
- Outputs the store's details and metrics
- Orders results by the performance category

```
select
  case
    when Top_Revenue_Rank = 1 then 'Top Revenue'
    when Bottom_Revenue_Rank = 1 then 'Bottom Revenue'
    when Top_Return_Rank = 1 then 'Top Return Rate'
    when Bottom_Return_Rank = 1 then 'Bottom Return Rate'
  end as Category,
  Store_ID,
  Store_Name,
  City,
  Country,
  Revenue,
  Return_Count,
  Total_Transactions,
  Return_Rate
from ranked
where Top_Revenue_Rank = 1
   or Bottom_Revenue_Rank = 1
   or Top_Return_Rank = 1
   or Bottom_Return_Rank = 1
order by Category;
```

Category	Store_ID	Store_Name	City	Country	Revenue	Return_Count	Total_Transactions	Return_Rate
Bottom Return Rate	23	Store Lyon	Lyon	France	10974699.516845703	6201	120968	0.05126
Bottom Revenue	18	Store Glasgow	Glasgow	United Kingdom	5117829.230078697	4326	81919	0.05281
Top Return Rate	27	Store Barcelona	Barcelona	España	7813191.347163439	4821	87482	0.05511
Top Revenue	6	Store 上海	上海	中国	300431731.3803978	20009	383212	0.05221

8. Segment products into price bands and analyze their contribution to total revenue and returns.

Explanation:

- We categorize products into price ranges (bands) based on Unit_Price. This allows us to group sales and returns by price segment. Result is Each row in the output corresponds to one price band.
- Only sums Line_Total for sales (ignores returns) gives total revenue generated by that price band.
- Sums Line_Total for returns shows how much value was returned in that price band.
- `sum(... = 'Return')` counts number of return transactions. `count(*)` is total transactions in that band. `Return_Rate` is fraction of transactions that were returns.
- Calculates percentage of total revenue that this price band contributes. Denominator is sum of all sales in all bands. Result for e.g., “Mid” band contributes 5.29% of total revenue.
- Calculates percentage of total return value attributable to this price band. Denominator is sum of all returns across all bands.
- Groups all transactions into their price bands. Orders the result so that highest revenue bands appear first.

```
Select
  case
    when t.Unit_Price < 50 then 'Low'
    when t.Unit_Price between 50 and 150 then 'Mid'
    when t.Unit_Price between 150 and 300 then 'Upper-Mid'
    else 'Premium'
  end as Price_Band,

  sum(case when t.Transaction_Type = 'Sale' then t.Line_Total else 0 end) as Revenue,
  sum(case when t.Transaction_Type = 'Return' then t.Line_Total else 0 end) as Return_Value,

  sum(case when t.Transaction_Type = 'Return' then 1 else 0 end) /
  count(*) as Return_Rate,

  sum(case when t.Transaction_Type = 'Sale' then t.Line_Total else 0 end) /
  (select sum(Line_Total) from transactions where Transaction_Type = 'Sale')
  as Revenue_Contribution,

  sum(case when t.Transaction_Type = 'Return' then t.Line_Total else 0 end) /
  (select sum(Line_Total) from transactions where Transaction_Type = 'Return')
  as Return_Contribution

from transactions t
group by Price_Band
order by Revenue desc;
```

	Price_Band	Revenue	Return_Value	Return_Rate	Revenue_Contribution	Return_Contribution
►	Premium	456791743.11569977	-25454216.30168152	0.0528	0.5886032465007487	0.5876448279837956
	Mid	125452513.89552498	-7025771.970287323	0.0529	0.161653002869369	0.16219939800936337
	Upper-Mid	104649751.24927521	-5821361.199806213	0.0527	0.1348474096985097	0.1343939550268408
	Low	89166514.39432836	-5014298.1601707935	0.0530	0.11489634093137259	0.11576181898000029

9. Calculate the average number of items and revenue per basket across different countries.

Explanation:

- Groups by Invoice_ID (one basket) sum(Quantity) is total items in that basket. Sum(Line_Total) is total revenue for that basket.
- Join stores on basket, Associates each basket with a country.
- avg(items_per_basket) Gives average basket size (items). avg(revenue_per_basket) Gives average revenue per basket.
- Grouped by country for comparison.
- Sorts countries by highest average revenue per basket.

```
Select
    s.Country,
    avg(items_per_basket) as Avg_Items_Per_Basket,
    avg(revenue_per_basket) as Avg_Revenue_Per_Basket
from (
    select
        t.Invoice_ID,
        t.Store_ID,
        sum(t.Quantity) as items_per_basket,
        sum(t.Line_Total) as revenue_per_basket
    from transactions t
    group by t.Invoice_ID, t.Store_ID
) basket
join stores s on basket.Store_ID = s.Store_ID
group by s.Country
order by Avg_Revenue_Per_Basket desc;
```

Country	Avg_Items_Per_Basket	Avg_Revenue_Per_Basket
中国	1.5555	490.13187220671426
United States	1.5546	62.821765780941234
España	1.5545	56.562261639205786
Portugal	1.5514	56.53964230478131
France	1.5561	56.346096309939156
Deutschland	1.5562	56.27125145449217
United Kingdom	1.5553	39.25094341445475

10. Analyze month-over-month revenue growth overall and for the top three categories.

Explanation:

Step 1 – Aggregate Monthly Revenue (monthly_overall CTE):

- Groups all transactions by month using `date_format(Date, '%Y-%m-01')`.
- Calculates total revenue per month by summing `Line_Total`.

Step 2 – Calculate Month-over-Month Growth:

- Uses `LAG()` to get the previous month's revenue.
- Computes:
 - `MoM_Change` = difference in revenue from previous month
 - `MoM_Growth` = percentage change from previous month

```
#Overall Month-over-Month Revenue Growth
with monthly_overall as (
  select
    date_format(Date, '%Y-%m-01') as Month,
    sum(Line_Total) as Revenue
  from transactions
  group by date_format(Date, '%Y-%m-01')
)
select
  Month,
  Revenue,
  Revenue - lag(Revenue) over (order by Month) as MoM_Change,
  (Revenue - lag(Revenue) over (order by Month)) / lag(Revenue) over (order by Month) as MoM_Growth
from monthly_overall
order by Month;
```

Continue on next page.

10. Analyze month-over-month revenue growth overall and for the top three categories.

Explanation:

Continue from previous page:

Step 3 – Output:

- Month – first day of the month
- Revenue – total revenue that month
- MoM_Change – increase/decrease vs previous month
- MoM_Growth – percentage growth vs previous month
- Ordered by month for a chronological view.

Top 3 Categories Month-over-Month Revenue Growth

Step 1 – Aggregate Monthly Revenue per Category (monthly_revenue CTE):

- Joins transactions with products using Product_ID to get the Category.
- Groups by month and category.
- Calculates total revenue per category per month.

Step 2 – Identify Top 3 Categories (top_categories CTE):

Continue on next page

Month	Revenue	MoM_Change	MoM_Growth
2023-01-01	13901921.780988693	NULL	NULL
2023-02-01	11344209.790790081	-2557711.990198612	-0.18398261984874367
2023-03-01	30245575.189388752	18901365.39859867	1.6661685341841923
2023-04-01	23409291.969956398	-6836283.219432354	-0.22602589557730648
2023-05-01	24079336.009550095	670044.0395936966	0.028622994683206757
2023-06-01	18467461.659989357	-5611874.349560738	-0.23305768677902974
2023-07-01	17215818.800148487	-1251642.85984087	-0.06777557646444797
2023-08-01	16166374.300314903	-1049444.4998335838	-0.06095815203541364
2023-09-01	34150378.278960705	17984003.9786458	1.1124327350440903
2023-10-01	35913489.01907539	1763110.7401146889	0.05162785389117937
2023-11-01	23425144.300333858	-12488344.718741536	-0.34773409824114754
2023-12-01	56636979.94991946	33211835.649585605	1.4177857444025335
2024-01-01	17051647.537905216	-39585332.412014246	-0.6989308477785553
2024-02-01	16081078.939182758	-970568.5987224579	-0.0569193443956027
2024-03-01	36553722.47589874	20472643.536715984	1.2730889273127595
2024-04-01	30998719.03989315	-5555003.436005592	-0.15196820076719184
2024-05-01	27095415.949486732	-3903303.090406418	-0.1259182060195179
2024-06-01	23760693.639995575	-3334722.3094911575	-0.12307330198244575
2024-07-01	21038678.399870396	-2722015.2401251793	-0.1145595865746656
2024-08-01	21009117.29942274	-29561.100447654724	-0.0014050835269118...
2024-09-01	44236522.52799988	23227405.228577137	1.105586917219761
2024-10-01	42959853.364607334	-1276669.1633925438	-0.02886007060307386
2024-11-01	31523706.39839065	-11436146.966216683	-0.26620544695896015
2024-12-01	70597973.20006514	39074266.801674485	1.2395200712714831
2025-01-01	14417731.72804737	-56180241.472017765	-0.7957769738347946
2025-02-01	13047083.098956585	-1370648.629090786	-0.09506687008362143

10. Analyze month-over-month revenue growth overall and for the top three categories.

Explanation:

Continue from previous page:

- Sums revenue for each category across all months.
- Orders by total revenue descending and selects top 3 categories.

Step 3 – Calculate Month-over-Month Growth for Top Categories:

- Uses LAG() with partition by Category to compare revenue within each category month-to-month.
- Computes:
 - MoM_Change – difference from previous month for that category
 - MoM_Growth – percentage growth from previous month for that category

Step 4 – Filter and Output:

- Only includes the top 3 categories.
- Columns:
 - Month – first day of month

Continue on next page

```
#Top 3 Categories Month-over-Month Revenue Growth
with monthly_revenue as (
    select
        date_format(t.Date, '%Y-%m-01') as Month,
        p.Category,
        sum(t.Line_Total) as Revenue
    from transactions t
    join products p on t.Product_ID = p.Product_ID
    group by date_format(t.Date, '%Y-%m-01'), p.Category
),
top_categories as (
    select Category
    from monthly_revenue
    group by Category
    order by sum(Revenue) desc
    limit 3
)
select
    Month,
    Category,
    Revenue,
    Revenue - lag(Revenue) over (partition by Category order by Month) as MoM_Change,
    (Revenue - lag(Revenue) over (partition by Category order by Month)) / lag(Revenue)
    over (partition by Category order by Month) as MoM_Growth
from monthly_revenue
where Category in (select Category from top_categories)
order by Category, Month;
```


10. Analyze month-over-month revenue growth overall and for the top three categories.

Explanation:

Continue from previous page:

- Category – product category
- Revenue – total revenue for that category that month
- MoM_Change – change vs previous month
- MoM_Growth – % change vs previous month
- Ordered by Category and Month.

Month	Category	Revenue	MoM_Change	MoM_Growth
2023-01-01	Children	1391900.4499840736	NULL	NULL
2023-02-01	Children	835450.9900064468	-556449.4599776268	-0.3997767656329113
2023-03-01	Children	3628070.368961811	2792619.378955364	3.342648955307139
2023-04-01	Children	1694579.1999702454	-1933491.1689915657	-0.5329254872046054
2023-05-01	Children	2501699.5802316666	807120.3802614212	0.47629546041612764
2023-06-01	Children	1359360.7899923325	-1142338.790239334	-0.45662508770679383
2023-07-01	Children	1249097.50026083	-110263.28973150253	-0.08111407254296667
2023-08-01	Children	1143463.6500897408	-105633.85017108917	-0.08456813831508851
2023-09-01	Children	3908170.465092659	2764706.8150029182	2.417835332838031
2023-10-01	Children	3425607.590803623	-482562.8742890358	-0.12347539049261883
2023-11-01	Children	1658362.3001312017	-1767245.2906724215	-0.515892507774902
2023-12-01	Children	4073053.049975872	2414690.7498446703	1.4560694907582206
2024-01-01	Children	1663649.228556633	-2409403.821419239	-0.591547370450653
2024-02-01	Children	1147178.310092926	-516470.91846370697	-0.310444599497571
2024-03-01	Children	4541808.948444843	3394630.6383519173	2.959113337905542
2024-04-01	Children	2283738.9098768234	-2258070.03856802	-0.49717415774197277
2024-05-01	Children	2819195.6886310577	535456.7787542343	0.23446497164735652
2024-06-01	Children	1799557.8900003433	-1019637.7986307144	-0.3616768437688088
2024-07-01	Children	1530432.850209713	-269125.03979063034	-0.1495506431252284
2024-08-01	Children	1563188.1501374245	32755.299927711487	0.021402637772198287

PART 2

POWER BI DASHBOARD

POWER BI INSIGHTS

- SALES & RETURNS OVERVIEW
- GEOGRAPHIC INSIGHTS
- CATEGORY & PRODUCT INSIGHTS
- PROMOTION EFFECTIVENESS
- CUSTOMER BEHAVIOUR

FOR WORKING DASHBOARD, PLEASE REFER BELOW LINK:

[Global Fashion Retail Analysis – Power BI](#)

Sales & Return Overview



Geographic Insights



○

○

Customer Behaviour

Total Revenue

65.08M

0.3bn

Children

Year-Month

Promotion Effectiveness



Customer Behaviour



All pages contains Filters as shown:



To open Filter pan



To clear all applied Filters



To close the filters pan

Geographic

Date

01-01-2023 18-03-2025

Country

All

Store Id

All

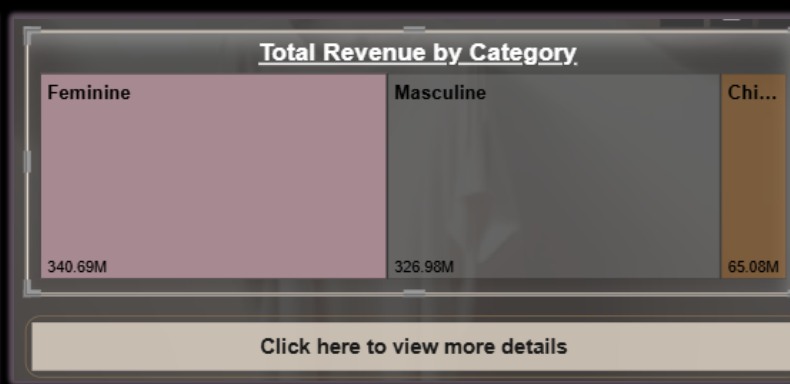
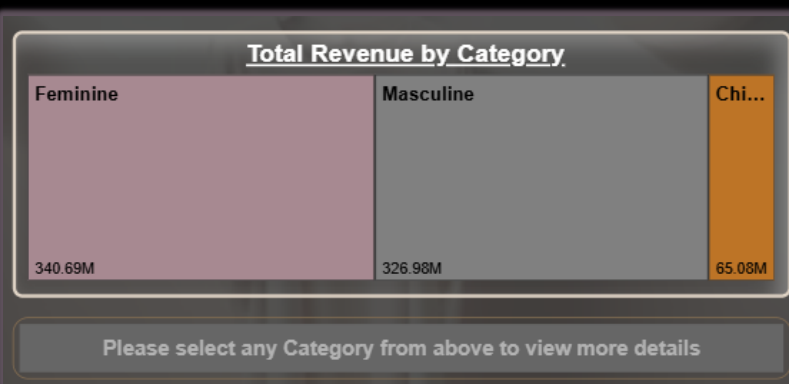
Category

All

Mode of Payment

All

Category & Product Insights page has Drill down option to view more details based on each category:



Global Fashion Retail Analysis

Color: All

Discounts: 0.20, 0.25, 0.35, 0.40, 0.45

340.69M Total Revenue

Product ID	Category	Sub Category	Country	City	Store ID	Store Name
17925	Feminine	Accessories	Deutschland	Berlin	11	Store Berlin
17925	Feminine	Accessories	Deutschland	Hamburg	12	Store Hamburg
17925	Feminine	Accessories	Deutschland	München	13	Store München
17925	Feminine	Accessories	España	Valencia	28	Store Valencia
17925	Feminine	Accessories	France	Marseille	22	Store Marseille
17925	Feminine	Accessories	Portugal	Braga	33	Store Braga
17925	Feminine	Accessories	United Kingdom	Birmingham	17	Store Birmingham
17925	Feminine	Accessories	United Kingdom	Liverpool	19	Store Liverpool
17925	Feminine	Accessories	United Kingdom	London	16	Store London
17925	Feminine	Accessories	United States	Chicago	3	Store Chicago
17925	Feminine	Accessories	United States	New York	1	Store New York
17925	Feminine	Accessories	中国	上海	6	Store 上海
17925	Feminine	Accessories	中国	北京	7	Store 北京
17925	Feminine	Accessories	中国	广州	8	Store 广州
17925	Feminine	Accessories	中国	深圳	9	Store 深圳
17925	Feminine	Accessories	中国	重庆	10	Store 重庆
17924	Feminine	Lingerie and Pajamas	Deutschland	Berlin	11	Store Berlin
17924	Feminine	Lingerie and Pajamas	Deutschland	Frankfurt am Main	15	Store Frankfurt am Main
17924	Feminine	Lingerie and Pajamas	Deutschland	Hamburg	12	Store Hamburg
17924	Feminine	Lingerie and Pajamas	Deutschland	Köln	14	Store Köln
17924	Feminine	Lingerie and Pajamas	Deutschland	München	13	Store München
17924	Feminine	Lingerie and Pajamas	España	Barcelona	27	Store Barcelona
17924	Feminine	Lingerie and Pajamas	España	Valencia	28	Store Valencia
17924	Feminine	Lingerie and Pajamas	France	Marseille	22	Store Marseille
17924	Feminine	Lingerie and Pajamas	France	Toulouse	24	Store Toulouse
17924	Feminine	Lingerie and Pajamas	Portugal	Coimbra	35	Store Coimbra
17924	Feminine	Lingerie and Pajamas	Portugal	Guimarães	34	Store Guimarães
17924	Feminine	Lingerie and Pajamas	Portugal	Lisboa	31	Store Lisboa
17924	Feminine	Lingerie and Pajamas	United Kingdom	Liverpool	19	Store Liverpool
17924	Feminine	Lingerie and Pajamas	United Kingdom	London	16	Store London

PART 3

MACHINE LEARNING – PREDICTIVE MODELLING

MACHINE LEARNING – PREDICTIVE MODELLING

- RETURN PREDICTION MODEL
- MONTHLY DEMAND FORECASTING

FOR CODE + EVALUATION OUTPUTS FOR RETURN PREDICTION AND FORECASTING PLEASE REFER BELOW LINK:

[Predictive_Model_ML.ipynb](#)

RETURN PREDICTION MODEL

Logistic Regression, Random Forest Model Result

Overall Model Performance

- Logistic Regression outperforms Random Forest across all key metrics.
- Higher ROC-AUC (0.94) indicates better ranking of return vs non-return cases.

Return Class (Minority Class) Performance

- Logistic Regression: Better precision (0.62) → fewer false positives
 - Better recall (0.26) → detects more actual returns
 - Random Forest: Struggles significantly with minority class detection
- Lower F1-score indicates poor balance between precision and recall

Since returns are the critical business class, Logistic Regression is superior.

Impact of Class Imbalance

- Dataset is highly imbalanced: ~95% non-returns, ~5% returns
- Logistic Regression: More stable decision boundary, Handles imbalance better without heavy tuning
- Random Forest: Biased toward majority class, Requires additional tuning (class weights, resampling)

RETURN PREDICTION MODEL

Logistic Regression, Random Forest Model Result

Why Logistic Regression Performs Better here

- Linear relationships between features and return behavior
- Strong signal from variables like:
 - Discount level
 - Category
 - Basket size
 - Payment method
- Probabilistic output allows threshold tuning

Model Selection Decision

Logistic Regression is selected as the final model for return prediction due to higher ROC-AUC, better minority-class performance, and stronger business interpretability. Logistic model can reliably Flag high-risk return orders and enable preventive actions (fit guidance, restricted returns)

Lower false positives ensures good customer experience

- Logistic Regression demonstrates superior predictive performance and business usability compared to Random Forest, making it the preferred model for return prediction in this retail dataset.
-

MONTHLY DEMAND FORECASTING

SARIMA Vs Gradient Boosting Model Result

Accuracy

- Gradient Boosting outperforms SARIMA
 - ~34% lower error in MAPE
 - ~20% lower error in RMSE

Reason

- GBR uses lagged sales + rolling averages + trend features, allowing it to capture local patterns even with volatile sales.
- SARIMA struggles because retail demand is:
 - non-stationary
 - highly variable
 - influenced by category/country effects

As a result, SARIMA's predictions fluctuate more wildly and deviate more from actual values.

Gradient Boosting is the better forecasting model for this series. It provides more stable predictions, lower error, and better alignment with actual sales

-
-

Summary Insights Report

EXECUTIVE SUMMARY

Strategic focus should shift from volume-driven growth to precision-led performance by improving promotion targeting, reducing returns through better product clarity, optimizing inventory around demand hotspots, and deepening engagement with a strong repeat customer base.

- Promotions are underperforming with a –97.47% lift in sales, contributing only ~2.5% of total sales.
- Feminine category dominates revenue (~340M+) but also has a high return rate (~5.6%).
- Repeat customers drive growth (75% of customers), while new customer acquisition remains low (~25%).
- Revenue volatility is visible across months, indicating inventory and demand planning gaps.
- Urban cities (Shanghai, Guangzhou, Shenzhen) generate the majority of revenue, signaling regional concentration risk.

Summary Insights Report

PROMOTION PERFORMANCE – KEY INSIGHTS

Insights

- Non-promoted sales significantly outperform promoted sales.
- Promotions may be **poorly targeted or over-discounted**.
- Promotional spikes do not sustain post-promotion growth.

Strategic Recommendations

- Shift from blanket discounts to targeted promotions
 - Use customer segments (repeat vs new) and category performance.
- Test value-based promotions
 - Bundles, loyalty points, free shipping instead of price cuts.
- Limit promotions to slow-moving SKUs
 - Avoid discounting top sellers (especially Feminine category).

Expected Impact

- Improved promo ROI
- Reduced margin erosion
- ➤ Higher conversion among targeted segments
-

Summary Insights Report

RETURNS ANALYSIS – KEY INSIGHTS

Insights

- Return value: 43.32M
- Categories with highest revenue also likely driving returns
- Online / card-based payments dominate → higher return propensity

Strategic Recommendations

- Category-specific return controls
 - Stricter return windows for high-return SKUs
 - Flag repeat returners and apply conditional free returns
- Pre-purchase nudges
 - “Low return rate” badges on best-performing products
 - Fit recommendations using past purchase behavior
- Post-return analytics
 - Track return reasons by category, city, and SKU
 - Feed insights into design, sourcing, and merchandising teams

Expected Impact

- 1–2% reduction in return rate
- ➤ Lower reverse logistics cost
- ➤ Improved customer satisfaction

Summary Insights Report

INVENTORY OPTIMIZATION – KEY INSIGHTS

Insights

- Monthly revenue spikes indicate reactive inventory management.
- Children category contributes least revenue (~65M), indicating potential overstock risk.
- High dependency on few categories and cities increases demand risk.

Strategic Recommendations

- Demand-driven inventory planning
 - Align inventory with historical monthly trends.
- Rationalize low-performing SKUs
 - Reduce assortment in Children category; focus on top-performing items.
- Regional inventory allocation
 - Prioritize high-revenue cities; avoid uniform distribution.

Expected Impact

- Lower holding costs
- Reduced stockouts and markdowns
- Improved inventory turnover

Summary Insights Report

CUSTOMER RETENTION & GROWTH – KEY INSIGHTS

Insights

- 75% repeat customers indicate strong loyalty.
- New customer growth is volatile and promotion-driven.
- Average basket size (~1.55 items) shows cross-sell opportunity.

Strategic Recommendations

- Strengthen loyalty programs
 - Tier-based rewards for repeat customers.
- Personalized recommendations
 - Cross-sell based on past purchases and category preference.
- Win-back campaigns
 - Target inactive customers with personalized offers.

Expected Impact

- Higher customer lifetime value (CLV)
- Increased basket size and frequency
- More stable revenue growth

Summary Insights Report

KEY STRATEGIC PRIORITIES

- Optimize promotions → Targeted, data-driven, value-based
- Reduce returns → Better product clarity + SKU-level action
- Streamline inventory → Demand forecasting + SKU rationalization
- Enhance retention → Loyalty, personalization, cross-sell

FINAL BUSINESS IMPACT

Implementing these strategies can:

- Increase promotion effectiveness by 2–3x
- Reduce returns by up to 20%
- Improve inventory turnover by 15–25%
- Drive sustainable, repeat-led revenue growth
- Strengthen customer lifetime value through retention



Thank You

Presented by: Shivani Awasthi